Rational Choice Overload

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Introduction: what's choice overload?

- Choice overload:
  - → "the paradoxical finding that providing individuals with more options can be detrimental to choice" (Chernev et al., 2015)

- More heavily studied in Psychology and Marketing, more lightly in Economics

- Detrimental? Various possible definitions.
  - → Some outside traditional scope of Economics: confidence*, satisfaction
  - → Others very much within it: choice

- This project: default choice
  - → not choosing one of the options presented by staying with a default choice
  - → choice overload: more default choice in a larger choice set

Introduction: the Classic Example

- Iyengar & Lepper (2000): supermarket jam display.
- Consumers faced either:
  - $\rightarrow$ 6 jams, or
  - $\rightarrow$ 24 jams.
- Many more people bought jam from the smaller display (12% vs. 2%)

Introduction: a troubled literature

- Literature on choice overload shows trouble and inconclusiveness.

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- Attempts at directly replicating the original results were unsuccessful (Scheibehenne et al., 2010, p.412).
  - → although similar findings in independent studies have been reported (Chernev et al., 2015, p. 335)

- Some meta-analyses, like Scheibehenne et al. (2010), conclude in favor of the non-existence of choice overload.

- Chernev et al. (2010), on the other hand, criticize their methods and conclusions.

Introduction: a troubled literature

- Dean, Ravindran and Stoye (WP) provide better tests and more solid evidence for CO
- Yet, no theoretical explanation has solidified itself as standard
- The puzzle of choice overload: missed good alternatives

Introduction: Main Idea

- Standard explanations say people do not really engage with large menus.
- Our alternative: people *do* begin to search, but may stop before they reach something good.
- Implications for Economic Design

Introduction: Other approaches

Currently two main classes of model used to explain choice overload:

- **Contextual strategic inference.** Kamenica (2008), Kuksov and Villas-Boas (2010), Nocke and Rey (2021)
  - $\rightarrow$ DM can make inferences about the nature of a set of alternatives based on its size, and the assumption that the set has been chosen by a profit maximizing firm.
- **Decision avoidance.** Beattie et al. (1994), Dean (2008), Gerasimou (2018)
  - $\rightarrow$ DM avoids engaging with large choice sets because they find it aversive to do so.

Call these two explanations "classic" choice overload.

Three Explanations to Compare

Classic

- People avoid engaging with large sets.
- Search often never starts.

Fatigue / increasing costs

- People start searching.
- Continuing becomes harder over time.

Learning

- People start searching.
- What they see changes what they expect to find next.

Standard Sequential Search

Standard sequential search framework:

- Decision maker faces finite set of alternatives X
- Has to sequentially search through X
- There's a *default option* $d \in X$, which starts out searched.

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- Decision maker faces finite set of alternatives X
- Has to sequentially search through X
- There's a *default option* $d \in X$, which starts out searched.
- Sequentially searching through X:
  - $\rightarrow$ In each period, can *stop* and pick the *best option seen so far* y, or *search* at cost k and look at a new option x drawn from a known distribution F.

Standard Search Does *Not* Generate Overload

- Solution: constant threshold τ over utility values such that:
  - $\rightarrow$ DM stops if $u(y) \geq \tau$
  - $\rightarrow$ searches otherwise
- Such constant thresholds cannot lead to choice overload. Why?
  - $\rightarrow$ If d not chosen in a small set, search started, i.e. $u(d) < \tau$, and something better got chosen
  - $\rightarrow$ Thus search also starts in the larger set, continues until better option is found

Standard sequential search alone cannot generate choice overload.

What has to change?

- To get overload inside a rational search framework, the stopping rule must be **non-constant**.
- Two natural ways this can happen:
  - $\rightarrow$ **Period dependence:** search gets harder the longer it goes on.
  - $\rightarrow$ **History dependence:** the values already seen change beliefs about what remains.
- These two possibilities lead to different behavioral predictions.

Theory Overview

Data

Threshold Functions

Optimal Behavior

Theory Overview

Data

Threshold Functions

Optimal Behavior

Models of optimizing agent

Theory Overview

Data

Description of observed behavior

Threshold Functions

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Representing solutions from optimal models

Optimal Behavior

Models of optimizing agent

Top Layer: Data

We consider two types of data

1. Standard stochastic choice data: $p(a, A)$
   - $\rightarrow$ A is a choice set (always contains d)
   - $\rightarrow$ $p(a, A)$ is the probability of choosing a from set A
   - $\rightarrow$ Allows us to determine which models generate choice overload

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2. Experimental choice data: $\rho(x, y, l)$
   - $\rightarrow$ l is a choice list (ordered set of elements in X, with d the first element)
   - $\rightarrow$ Probability that search stops at y and x is chosen from list l
     - x as chosen option, choice point
     - y as stopped option, stopping point
   - $\rightarrow$ Allows us to differentiate between causes of choice overload
   - $\rightarrow$ Will collect such data in our experiment

Top Layer: Data

- Given standard choice data: $p(a, A)$:

Definition

We say that a data set **exhibits choice overload** if, for some A, B such that $A \subset B$, $p(d, B) > p(d, A)$.

Theory Overview

Data

$p(a, A)$, $\rho(x, y, I)$

Threshold Functions

Solution

Optimal Behavior

Model of optimizing agent

Theory Overview

Data

Data that cannot exhibit choice overload

Threshold Functions

Constant threshold

Optimal Behavior

Standard sequential search

Theory Overview

Data

$p(a, A)$ (test CO), $\rho(x, y, I)$ (check search behavior)

Threshold Functions

Solutions

Optimal Behavior

Search-based	**Classic**
Learning Increasing Costs Static Search	Decision Avoidance Contextual Inference

Increasing Costs and Learning

Increasing costs

- Search threshold falls as search continues.
- Later good options are more likely to be missed.
- What matters is **how late** the first good option appears.

Learning

- Search threshold changes because beliefs change.
- Early bad draws can make the person pessimistic.
- So not only timing but also the **values seen along the way** matter.

What the experiment has to identify

- **Classic vs. search-based:**
  - $\rightarrow$ if classic models are right, order should not matter much because search often never starts;
  - $\rightarrow$ if search models are right, order should matter.
- **Increasing costs vs. learning:**
  - $\rightarrow$ if only fatigue matters, the position of the first good option should do the work;
  - $\rightarrow$ if learning matters, the values seen before that good option should matter too.

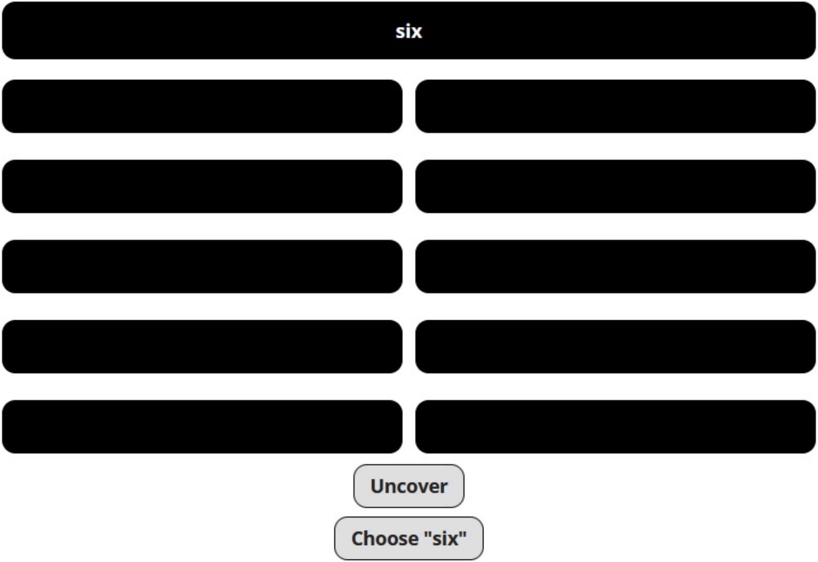
Experimental Task: what it is designed to do

- Give subjects options with clear monetary value, but values that take effort to process.
- Control the **order** in which options are uncovered.
- Make the decision to **stop searching** directly observable.
- Control beliefs about the distributions from which values are drawn.

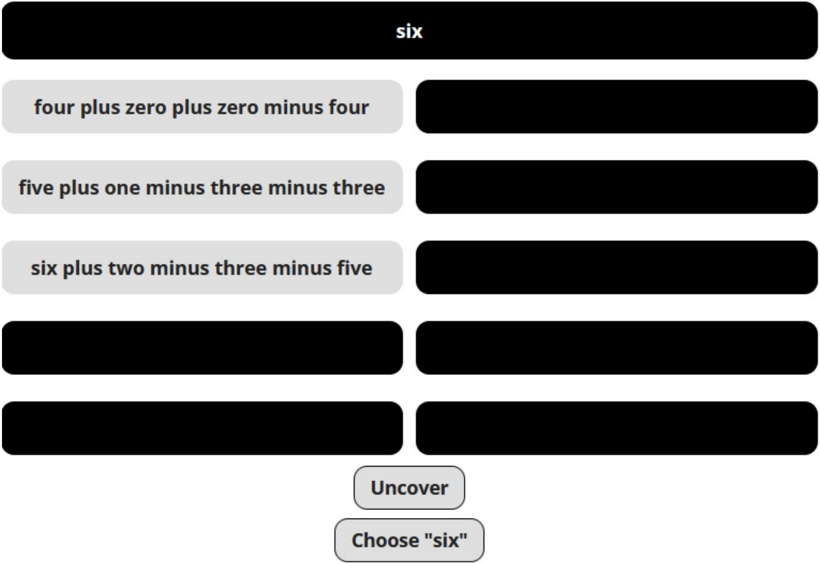
Task Overview

- Subjects see a default option and a set of covered alternatives.
- Non-default options must be uncovered sequentially.
- The uncovered value is a monetary payoff displayed as a verbal arithmetic expression.
- At any point the subject can stop and choose the best option seen so far.

Experimental Task



Experimental Task



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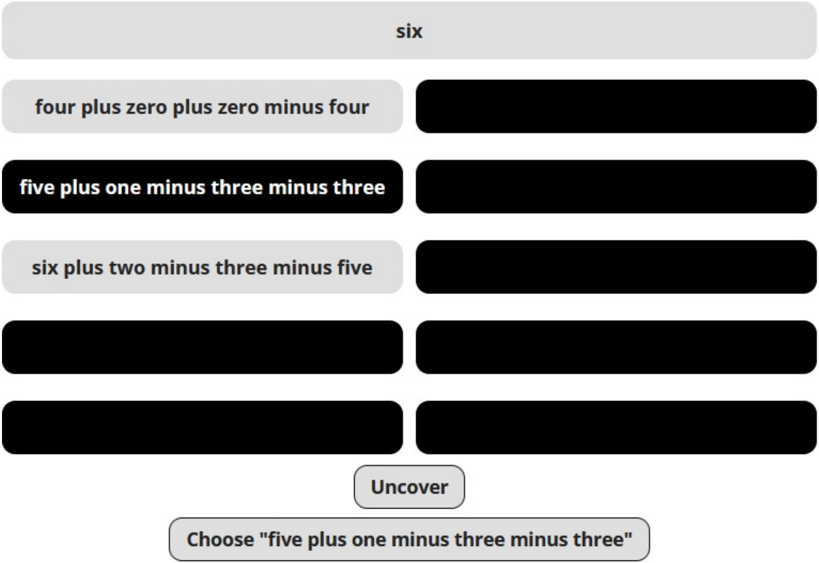
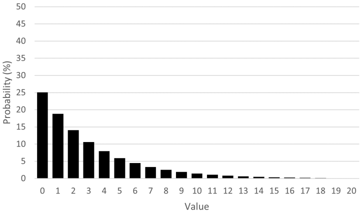
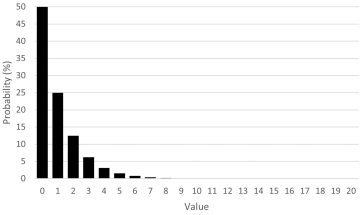


Experimental Setup

- Each round, options are drawn from one of two distributions.
- Default value is 6 points.
- Set sizes: default and 1,10,15,20.





Experimental Setup

- Extensive instructions with practice in drawing from the distributions to understand them and how choice sets were being constructed
- 5 comprehension questions, subjects who failed the same question twice were routed out
- 307 subjects on Prolific
- 10 choices each

Results

- Question 1: Do we find choice overload?

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- Yes!

Set Size	**% Default Choice**	*N*
2	**9**	*53*
11	**38**	*211*
16	**41**	*273*
21	**44**	*274*

- Looking at sets in which there was a better option than default.

Results

OLS regression of default choice on set size dummies:

	Default Choice
S=11	0.285***
	(0.050)
S=16	0.316***
	(0.050)
S=21	0.347***
	(0.052)
Constant	0.094***
	(0.041)
R^2	0.028
N	811

Results

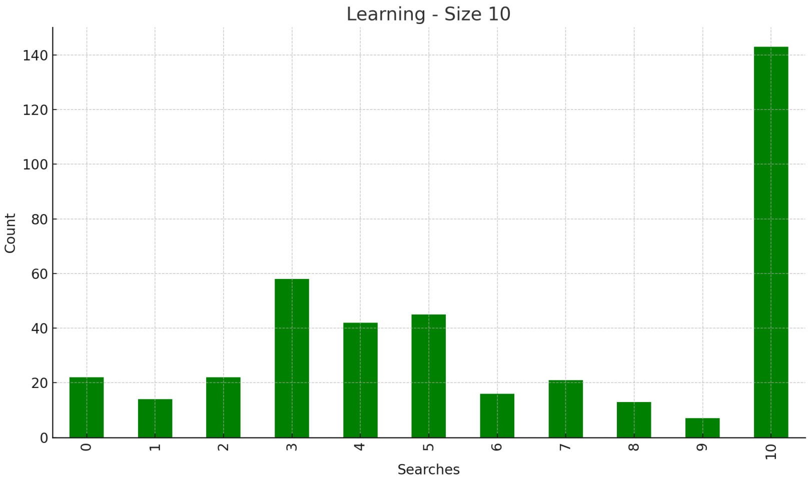
- Question 2: Is the increase in missing good options all due to never searching?

Size	**Choose Default**	**No search**	*N*
2	9%	2%	*53*
11	38%	7%	*211*
16	41%	5%	*273*
21	44%	5%	*274*
$s > 2$ **minus** $s = 2$	32%	4%	

Results

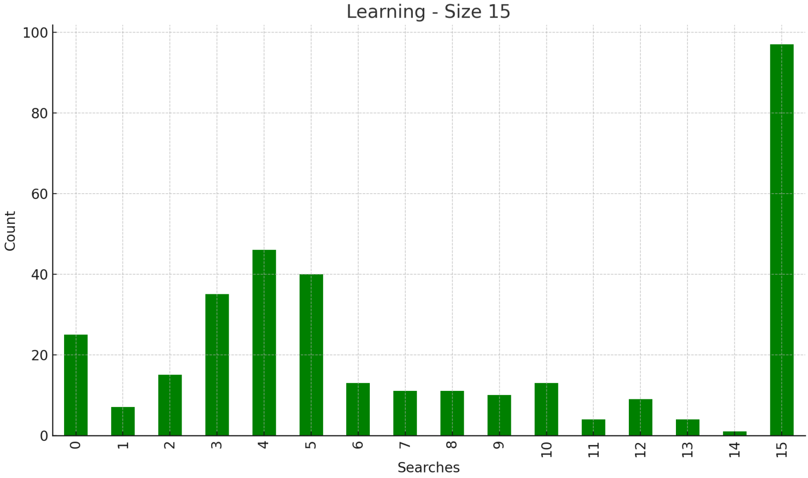
- Question 2.5: Is everyone standard or classic?
- Can look at sets *without* an option better than default.
- Under standard/classic CO models: search behavior completely on the extremes, i.e., no search or full search

Results



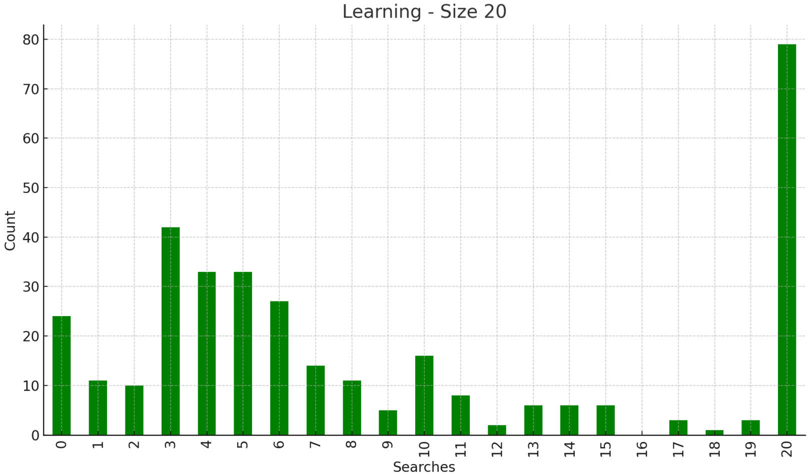
- Sets without better-than-default options
- Search distribution for size 10: 59% non-extreme

Results



- Sets without better-than-default options
- Search distribution for size 15: 64% non-extreme

Results

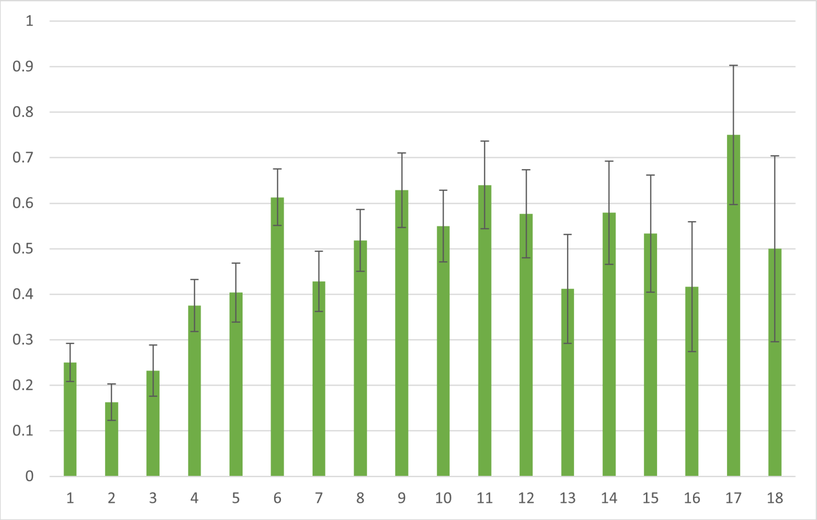


- Sets without better-than-default options
- Search distribution for size 20: 70% non-extreme

Results

- Question 3: Does search order affect choice?
  - $\rightarrow$ True in search models, not is classic models of overload
- Here we can use random variation in the position of the 1st above-default option in the search sequence
- Are people more likely to choose the default when better options occur later?

Results



- Probability of default choice by position of first better option
- Significant at <0.1%, robust to separate set sizes

Results

- Choice deferral may happen more often when first good option occurs later because
  - → Higher search costs
  - → Lower beliefs
- This leads to:
- Question 4: Does learning play a role in default choice?
  - → Can fix the position of the first above default option
  - → According to increasing costs: values before then should have no effect
- Again test this using random variation in search sequences

Results

- For each list position $k = 1, ..., 15$ look at sets in which the first above default option occurs after k
- Regress

$$chose\_default_{i,j} = \alpha + \beta min\_cost^{k}_{i,j} + \varepsilon_{i,j}$$

- Where
  - $\rightarrow$ i subject, j round
  - $\rightarrow$ $min\_cost^{k}_{i,j}$: lowest cost such that an optimal decision maker would stop searching at or before position k in set (i, j)
- Include also regression using beliefs

Results

	Min. Beliefs			Min. Costs		
k	**Coeff**	**s.e.**	**N**	**Coeff**	**s.e.**	**N**
1	-0.14	0.10	650	0.61	0.67	650
2	-0.37*	0.20	564	-0.67	0.48	564
3	-0.51	0.32	508	-0.59	0.40	508
4	-0.61	0.45	436	-0.44	0.39	436
5	-0.35	0.60	379	0.00	0.38	379
6	-0.49	0.80	317	0.04	0.39	317
7	-1.37	1.06	261	-0.29	0.38	261
8	-2.92**	1.22	207	-0.59	0.39	207
9	-3.58**	1.50	172	-0.84**	0.40	172
10	-3.85*	1.97	132	-1.14***	0.43	132
11	-5.22**	2.32	107	-1.65***	0.46	107
12	-8.99***	2.50	81	-1.78***	0.51	81
13	-8.10*	4.50	64	-1.66***	0.54	64
14	-11.37*	6.07	45	-1.98***	0.58	45
15	-11.03	6.82	30	-1.61**	0.72	30

Results

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- Method: Run a 'behavioral' regression
  - → Observation is behavior at list item k in list (i, j)
  - → Regress whether or not search has stopped by k

$$stop_{i,j}^{k} = \alpha_i + \beta_1 above_{i,j}^{k} + \beta_2 k + \beta_3 min\_cost_{i,j}^{k} + \varepsilon_{i,j}^{k}$$

- Where
  - → $min\_cost_{i,j}^{k}$ minimum cost at position k for set (i, j)
  - → k is number of items seen
  - → $above_{i,j}^{k}$ dummy for whether best thing seen is better than default
- Regression with beliefs and items remaining also

Results

	Searched Stopped	
	(1)	**(2)**
Min. Cost	-0.713***	
	(0.072)	
Min. Beliefs		-0.397***
		(0.041)
k	0.031***	0.029***
	(0.001)	*(0.002)*
Items Remaining		-0.007***
		(0.002)
Above	0.060***	0.084***
	(0.017)	*(0.017)*
Const	0.329***	0.322***
	(0.018)	*(0.026)*
R^2	0.542	0.537
N	24,182	24,182
Subject f.e.	Yes	Yes

Conclusion

- Theoretically-based experimental exploration of search mechanisms of choice overload.
- General family of models capturing optimal solutions of sequential search models.
  $\rightarrow$ Captures effects of limited information processing.
- Experiment designed to test these different models.
- Results show search channel is important
  $\rightarrow$ Highlight the role of learning
- Implications for Economic Design